

REST, INJURY PREVENTION, AND REHABILITATION



Jonathan Siegrist drinking in the fourth ascent of *Moonshine 5.14d*, Wild Iris, WY.

📷 Mike Anderson

CHAPTER 9

"I think that one of the most important issues in one's training plan is to know when you need to take a rest." — Iker Pou⁶

It's been four months of hard work and self-denial: honing the body, learning new skills, and applying the right strategies and tactics to goal routes. This work has produced some impressive results, and maybe even a new personal best. The new strength and fitness that have been diligently created have boosted confidence and magnified the enjoyment of climbing's many pleasures. After several weeks of riding this performance peak it can be hard to say goodbye to a magical season and start over with the hard work to prepare for the next season. Your newfound level of mastery is well-deserved, and it is tempting to try to milk it for as long as possible. The reality is that doing so simply isn't possible. The body will eventually fall off the peak. Even worse, attempting to extend a performance peak can lead to serious injuries, impacting future climbing seasons.

The final phase of the periodic training program is Rest. This chapter will discuss the importance of rest — within a training phase, during performance, and between seasons — and how to get the most out of prescribed downtime. This is the final piece of the puzzle that will enable each climber to build his or her own custom training plan. Finally, a brief discussion on injury prevention is also presented, and though hopefully it isn't needed, a few words of advice on rehabilitation.

"Rest is very important....Only when I have given my body the right amount of food and rest do I feel authorized to use it to do what I want to do."

— Fred Nicole⁷

REST

REST DAYS

Rest days are critical to the recovery process, facilitating progression as well as successful performance. They provide the body the opportunity to adapt to training stress, thus enabling improvement.

Rest days are not days in which nothing happens. During training, rest days include any days that don't have specific climbing workouts planned, though other supporting activities should take place. The optional Base-Fitness training days that follow intense workouts (during the Strength and PE phases) are also rest days. This program incorporates many rest days because it is focused on *improvement*. Though improvement is stimulated by exercise, that improvement *occurs* while resting. Therefore, climbers using this program may feel like they are resting too much and are thus "not tired enough," but that is not the mark of an effective training program. There are infinite, useless ways to get tired that will do nothing to improve a climber's abilities. If long-term improvement at climbing is the goal, appropriate rest is essential.

First and foremost, a productive rest day should do no further damage. Don't engage in activity that will further tax the muscles being trained, overstress weary joints, or place further wear and tear on the skin (sheltering finger skin is especially critical during the Performance Phase). It is acceptable to perform optional aerobic exercise on rest days, so long as the volume is limited to ensure it does not interfere with sport-specific training.

During the Performance Phase, rest days are those not spent attempting limit routes at the crag, and should be even more restful than rest days during training phases. Avoid long bouts of manual labor, or other taxing activities such as endurance sports. Climbers should try to stay off their feet and

out of harsh weather. Light activities, such as a hike to the crag to gawk at routes or belay a friend, are good Performance Phase rest-day activities. Set aside some time for quiet reflection to consider the previous climbing day, visualize the current project, and set short-term goals for the next day (see **Chapter 2: Goal Setting and Planning** for more details). Finally, perform any rehabilitation tasks such as skin care, stretching, massage, or icing.

Most climbers love being active, especially when visiting the outdoor playgrounds in which the great crags often reside. Climbers naturally want to explore with long hikes, bike rides, and so on, but this can be detrimental to climbing performance. Many sacrifices were made along the way to reach the performance peak, so the focus should be on maximizing climbing performance to the greatest extent possible. That said, it may be difficult to give up “rest-day activities,” especially when visiting a new part of the world. Therefore, this tradeoff should be considered early in the seasonal planning process. If non-climbing activities are very important to you, make a decision to include them and accept that climbing performance will be less than maximal. Alternatively, consider scheduling particularly grueling rest-day activities early in the trip, making it possible to rest thoroughly later in the trip, when a redpoint is more likely. Any such tradeoffs should be discussed with traveling companions well in advance to establish expectations and avoid disappointment or potential conflicts.

SLEEP

Sufficient sleep is vital to recovery and athletic performance. Research shows that improving the quantity and quality of sleep directly improves reaction time, power, endurance, cognitive function, technical skills, mood, and qualitative energy levels. Lack of sleep can lead to ineffective recovery, reduced performance, and general feelings of lethargy or poor motivation. Many climbers exacerbate this problem by carrying “sleep debt” for weeks on end, making it very difficult to perform effectively. Adults should attempt to average seven to eight hours of sleep per night; teens and young adults should average nine or more.¹ Several tips for improving sleep include:

- Account for sleep in daily planning
- Minimize sleep debt by getting regular sleep for several weeks prior to important performances
- Go to bed and wake up at regular times
- Attempt to get extra sleep (up to 10 hours per night) during the week leading up to important performances

“Good strong skin is critical.”

— Jerry Moffatt¹¹

SKIN CARE

Skin care is an essential rest-day activity, demanding attention during both the training and performance phases. Skin on the hands and fingers is the primary interface between the rock and the climber, and many climbing trips (and even entire seasons) have been derailed by skin issues: (toe and heel skin is also subject to abuse from climbing, and may warrant similar attention for some climbers). Acute problems like bloody flappers or splits in the pads can be disastrous to project climbers. Even onsite performance can be hampered by worn-down finger pads that ooze moisture.

Every performance-oriented climber should have a simple skin-care kit. This kit should come along on every trip to the crag or training facility, and should be used regularly (daily during the Performance Phase of the season). At a minimum, this kit should have:

- Two rolls of athletic tape
 - Wide roll for taping wide swaths
 - Roll of narrow strips for finger taping
- Nail clippers
- Neosporin or similar antibacterial ointment
- Band-aids (variety of sizes)
- Tweezers
- Needle
- Cuticle cutters (found in the beauty department)
- Sanding block (found at a hardware store)



Assemble a Skin Care Kit such as this, and keep it handy when climbing or training.

The best way to manage skin is to keep it healthy. First, prepare the skin for battle well in advance. Ideally, each climber should begin preparing the skin for the upcoming season when she begins her Strength Phase. In these initial stages, cut away any rough areas with cuticle cutters, then sand the finger pads the day after each training session. Keep these tools handy during the workout in the event of an issue. Repeated climbing can cause *calluses* (hard layers of dead skin) to form on the fingers and palms. These can lead to bloody *splits* or *flappers*. Sand calluses down between climbing sessions and apply lotion to prevent these injuries.

Topical skin balms can be applied after each training session. Be cautious during non-climbing activities to avoid unnecessary wear and tear. Get into the habit of using leather gloves for any type of construction, maintenance, or landscaping activity. (Look before you leap: Don't shove your hand into a closed box and hope there aren't open scissors waiting to greet you!) Keep the hands dry in the few hours preceding a workout or crag session; moist skin is soft and more susceptible to damage. Avoid extended hot tub soaks, baths, or showers prior to climbing.

Once on the rock (or at the gym), realize that skin quantity is limited, so use your skin judiciously. Avoid flailing on textured holds. Instead, once the fingers are placed on a hold, don't allow them to slip or slide. Avoid blind dynos; the target hold might have a razor-sharp lip. Carry the narrow-strip tape roll on climbs (clipped to a carabiner) so that tape can be preemptively applied when a threatening hold is discovered. Better to tape in the early stages of working a route, with the option of removing the tape for a committed redpoint attempt, than to forgo taping early and be forced to tape over a bloody flapper when it's time to send. Most acute injuries like lacerations and minor flappers begin as abrasions. Caution is the better part of valor in these cases, so quit early and save some skin for the next climbing day.

Chalk can be a great ally in protecting a climber's skin. Extra chalk should be carried to the crag or gym in the event of a massive spill, and the chalk bag should be inventoried before each roped burn to ensure there is an adequate supply. Thoroughly coat every part of each hand before starting up a climb, and continue to chalk regularly, especially on warm or humid days. Chalk is used as a dry lubricant in gymnastics and weightlifting, so thick layers of chalk can be used to alleviate hot spots: (areas of weakened/abraded skin).



Blind dynos to unknown holds can have serious consequences.

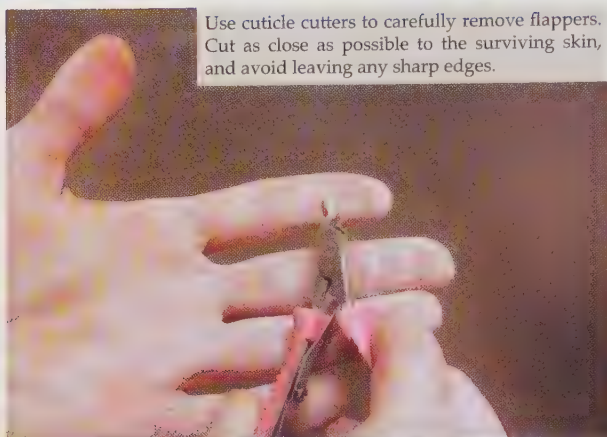
Even with the best preventative measures, skin injuries are inevitable. Any emerging problem areas should be dealt with swiftly. Common climber skin injuries include abrasions, "flappers" (which can be caused by minor lacerations or shear stress), and blisters (including blood blisters and water blisters).

ABRASIONS

Most climbing abrasions are minor and can be left alone to some extent. Serious skin issues are more often caused by repetitively using a single rough hold than by using a particularly sharp hold once. Minor abrasions become problematic when the abuse is continued, resulting in a more serious injury. Tape abraded areas early in a climbing session if the intent is to continue climbing. At the end of the day, clean the affected area, then apply a thin layer of Neosporin. Severe abrasions (those that are oozing blood or other moisture) should be treated immediately with Neosporin, and then covered with a band-aid and several wraps of tape. Depending on the location and effectiveness of taping, it may be possible to continue climbing.

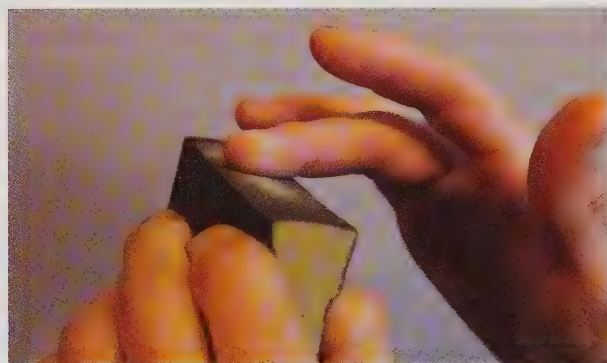
FLAPPERS

Flappers should be taped at the first sign of distress, particularly if you're still on the route. If the "flap" is minor (not oozing blood) and in an area that can't easily be taped, like a finger pad, simply continue to try other moves and save the sharp holds for the next burn. Once on the ground, first use cuticle cutters to remove any flaps of skin that have separated from the underlying layer. If not removed, these flaps can catch on sharp rock protrusions, thus causing further damage to the surviving tissue. If the newly exposed patch of skin is not tender, carefully sand the edges of the top layer of skin until they are flush with the newly exposed layer, to prevent these edges from catching on the rock. Ideally, this should be done without irritating



Use cuticle cutters to carefully remove flappers. Cut as close as possible to the surviving skin, and avoid leaving any sharp edges.

the newly exposed skin. If the skin is raw or sensitive, apply Neosporin, a Band Aid of the smallest size that covers the affected area, and then wrap the entire area with tape. If done correctly, it should be possible to continue climbing on all but the ugliest wounds. The affected areas should be washed and sanded every day. Raw skin should be treated with Neosporin and wrapped until it is no longer raw.

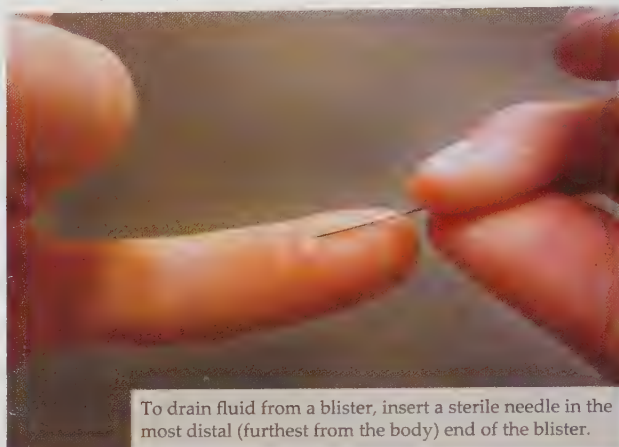


Once the flapper is trimmed away, use a sanding block or emery board to smooth the edges of the remaining outer layer of skin.

BLISTERS

Blisters are usually acute and can be slow to heal. A blister is basically a shear stress-caused flapper that hasn't opened up (yet). Eventually, the outer layer of skin will need to be removed, but in most cases the best bet is to keep the skin intact while draining any fluid with a sterilized needle. If a blister occurs mid-route, it may be possible to continue climbing with thorough application of tape. Once on the ground, use a lighter to sterilize a needle, let the needle cool, and then poke a tiny hole in the most distal (closest to the finger tip) end of the blister. The needle should be parallel to the finger when making the insertion. Remove the needle, drain the fluid (it may be necessary to reinsert the needle a few times to remove all fluid), and then re-tape the blistered area. It is often possible

to continue climbing once the blister is treated. Re-drain the blister periodically throughout the session and over the next few days. Eventually, the outer layer of skin will dry up and die. It's generally best to preserve this outer layer for as long as possible, since the underlying layer will be weak and sensitive. Once the blister opens up, treat as for a flapper.



To drain fluid from a blister, insert a sterile needle in the most distal (furthest from the body) end of the blister.

GERMAN FOOSBALL CREAM

The blogosphere is abuzz with whisperings of mystery ointments from distant lands that will transform even the hands of a princess into callused leather baseball mitts overnight. Some of these creams are designed for farm animals, some for industrial applications. The notorious *Anti-hydral* (sold at www.foosball.com) is labeled entirely in German, making it difficult to discern exactly what's in it or its intended use.

In general, these creams do in fact work. Some of them work so well it's scary. Be warned, however: Excessive use of these creams will *cause* far more problems than it prevents, and that is to say nothing of the unknown long-term side effects. Even modest use can dry and toughen the skin to the point of cracking, resulting in a bloody mess that requires weeks to recover from. Even without acute side effects, these creams often create a glassy, smooth surface on the skin that reduces tactile sensitivity to the rock and can make friction grips (like slopers and pinches) more difficult.

I've experimented with these products and my conclusion is that they work, but they are often not worth the risk. Diligent use of a sanding block and cuticle cutters will produce thick, yet still supple skin on the pads without the potentially catastrophic consequences. Plus, your significant other won't recoil at your touch.

— Mark Anderson

“You can take it for granted that people will train; it’s when and how much they rest that’s variable. Too much training is as bad as not enough.”

— Jerry Moffatt¹¹

THE FINAL PHASE

At the conclusion of every performance peak, it is essential to rest briefly before beginning training activities for the next season. This rest period is the final phase in the Rock Prodigy program, and it is just as important as the other phases. The periodic training approach works best when used in its entirety, and the Rest Phase is no different. This Rest Phase should not include any climbing whatsoever, and can be as short as five days or as long as four weeks, but typically two weeks are planned.

Rest plays an important part in physical training by allowing super-compensation to take place after training. This effect is clear for short rest periods following individual workouts, but it also applies to entire seasons. Certain desirable adaptations to training, such as the reinforcement of bone, tendon, and ligaments, occur on much longer time scales than adaptations within the muscles. Therefore, a longer, less frequent rest period is needed to allow those structures to adapt.

Furthermore, the motor processes that are used to control a climber’s movement and to develop new skills need rest as well. A brief rest period allows the central nervous system to reinforce the optimal neural networks for controlling movement, and clean up any less efficient commands that are leftover from the skill-acquisition activities. The improvement in skill execution that often accompanies a break from intense practice is known as the *reminiscence effect*, and is a result of this neural “housekeeping” process.

Time away from the all-encompassing pursuit of mastery over the complex sport of climbing also provides an opportunity to gain a new perspective. When mired in the details of daily workouts or cruxes, it is difficult to recognize the bigger picture and identify weaknesses in the larger process. A break from this minutia, especially if engaging in other, unrelated activities, may offer unusual insights about the overall process of climbing improvement that would be impossible to gain otherwise. Finally, resting restocks the motivational stores that are essential to the success of the coming training cycle. Use this time to enjoy other hobbies, spend time with neglected friends or family, or take a vacation to a non-climbing destination.

On the other hand, refusing to rest will lead to diminishing, and eventually evaporating, returns. Performance will stagnate, and then regress. While it is tempting to cling to the last drop of better-than-normal performance that characterizes the waning days of a performance peak, doing so stifles long-term progress. This is because sub-optimal performance days replace what would otherwise be productive improvement time. While performance climbing days are absolutely essential (they are the whole reason for training in the first place), performance days at the crag are not the most effective training for climbing. Performance climbing requires *avoiding* fatigue, thus the physical stimulus is not sufficient to trigger physiological improvements. Performance climbing utilizes well-mastered movements and skills in an urgent situation, thus it does not afford the time, motivation, setting, or novel sequences needed for effective skill development. Experienced climbers know that they get out of shape at the crag, and they get in shape with training — take it from Adam Ondra: *“You don’t train while trying a hard route; you need 100 percent rest before every single try, hence you need to rest a lot in general, and within a longer period of time some sort of regress is bound to take place.”*⁵

This is especially so for the last days of a Performance Phase when the climber typically makes fewer and fewer attempts on easier and easier routes, with more rest; call it garbage time. Trading effective training days for such garbage time is shortsighted. The climber may get lucky and add one or two more ticks to his list, but it’s at the expense of future progress. Recall the discussion of the Time Value of Climbing Ability from [Chapter 1: Introduction](#). That additional 5.1X redpoint is trivial compared to the value of the extra time it takes to achieve it, for as the climber improves, that route becomes easier and easier to “tick.” Save it for a future season, and focus near-term efforts on long-term improvement.

The optimal strategy for climbers seeking long-term, lasting progress is to complete as many macro-cycles as possible in his or her career. Tacking on four weeks of garbage time at the end of every season adds up to one fewer training cycle per year, and many lost opportunities for improvement. Instead of falling into this trap, be grateful for the season’s sends. For those routes that remain undone, accept that prioritizing long-term improvement is the surest way to accomplish them. Move on to the next macro-cycle, make new goals, and hatch a new plan to achieve them.

WHEN TO CALL IT QUILTS

It can be hard to decide when to end the Performance Phase, especially for those climbers who are new to periodic training (or any structured training, for that matter). Climbers experienced with periodic training can *feel* when their body is tiring and their performance is declining. They may also rely on past experience and simply terminate the Performance Phase after a given time period (say four weeks), based on the experience of previous seasons that went too long. Making this decision is more difficult for those using a structured training program for the first time. Often, the amount of improvement within the first few seasons of training is so great that even when performance declines, it is still higher than the previous baseline. This tempts the climber to extend the Performance Phase.

The Performance Phase should last at least three weeks (not including the Power-Endurance Phase), and in that time, every effort should be made to reap the benefits of the season's training. There are few long-term downsides to ending the phase too soon, however. Overextending the Performance Phase is much more problematic. To simplify the decision, start with this hard-and-fast rule: *Do not continue climbing beyond a six-week Performance Phase.* In general, the actual performance peak will be shorter than this. While this may seem like a brief payoff for months of hard work, keep in mind that the two-to-four week Power-Endurance Phase includes performance days as well, which provide ample sending opportunities, effectively lengthening the Performance Phase.

Besides temporal limits, physical indicators should be used to establish the end of the Performance Phase. Any serious injury, regardless of when it occurs, should end the season. These include:

- Finger-tendon or ligament tears
- Forearm muscle pulls
- Acute elbow or shoulder injuries
- Chronic upper-body aches or pains lasting more than two weeks
- Major bodily injuries (back, ankles, knees, broken bones, etc.)



When possible, end your season on a high note, such as after achieving a new personal best redpoint or onsight. Kristin Yurdin ready for a well-earned break after redpointing her first 5.14a, *Chemical Ali*, Smith Rock, OR.  Mike Anderson

Tendonitis or tendonosis may or may not be season ending depending on several factors, including the severity of the inflammation, the extent to which it is aggravated by current climbing activities, prevention/rehabilitation actions, and your pain tolerance. A minor injury early in a Performance Phase might be worth climbing through, whereas the same injury near the end of the phase is likely a sign to cease climbing. A more complete discussion of injury prevention and rehabilitation is presented later in this chapter.

Besides injury, a decline in physical performance is a sure sign to end the season, but often such declines are not obvious. If the climber is on the verge of sending a difficult project, his growing familiarity with the moves may mask any decline in physical fitness. Therefore, if continuous steady progress is not being made on the project, it is likely time to move

on and return to the project another season. Climbers shouldn't expect to reach a new high point on every go, but if three or more climbing days pass without additional linkage or a new high point, the route is probably unattainable *that season*. Power gradually decreases towards the end of the performance peak, and many experienced climbers use this as their cue to proceed to the Rest Phase. (Be very cautious about attempting "tweaky" or powerful routes after this power decline: Doing so often leads to injury.) Another indication of the end of the performance peak may be the need for longer and longer rest periods. If more than 48 hours of recovery is required between cragging days (which should be relatively low intensity, as explained previously), the peak has ended and so too should the Performance Phase.

Changing motivation or attitude can also warrant ending the season. Trying hard day after day on limit routes can be very mentally demanding, and climbers shouldn't feel ashamed or somehow less than committed if they lose motivation for this over several weeks. A lack of tangible success further exacerbates this problem. In such situations, it is critical to focus on the larger goal of long-term improvement over the career. The climber should embrace the learning opportunities that exist amidst the struggle of a hard project, but not affix his or her self-worth to the ultimate result. Doing so can lead to self-esteem injuries that are every bit as debilitating as physical ones. If necessary, move on to the next season, look forward to improving, and plan to come back another season, stronger and wiser.

REST PHASE ACTIVITIES

The Rest Phase should not be a time for all-out sloth and gluttony. While a break from climbing is necessary, other physical activities should be performed instead, making this a period of *active rest*. During the final phases of the season, non-climbing activities were reduced, so this is the time to pick them back up again. Go cycling, skiing, play tennis, or whatever it is that distracts from climbing and provides moderate exercise.

This is also a time for reflection, goal-setting, and planning. Review the last season and consider these questions:

- Were the goals appropriate (too easy or too hard)?
- Were the goals accomplished? If not, why?
- Were weaknesses improved? If not, why?
- How should the training plan be adjusted to:
 - Improve the chances of accomplishing the goals?

- Increase the improvement of weaknesses?
- Did the performance peak occur at the right (expected) time?
 - How long did the peak last?
 - Can this timing be predicted, to aid the planning of trips, vacation time, etc. in the future?
- Did the season reveal any previously unknown strengths and/or weaknesses?
- Does the season's improvement warrant re-prioritizing your weaknesses?

Reviewing the previous season should occur simultaneously with planning the next (discussed extensively in [Chapter 2: Goal Setting and Planning](#)). Each climber should consider the progress that was made toward his overarching and long-term goals, and decide if any course corrections are required. Determine what the intermediate and short-term goals should be for the next season and build that plan, then get psyched for the next season!

INJURY PREVENTION AND REHABILITATION

Injuries are not a pleasant topic, but given the unusual and extreme loads that we climbers place on our bodies, they are likely to happen at some point in every career. The fingers and forearm muscles are the most susceptible to injury in climbing, and these injuries are also the most debilitating. Therefore, finger injuries (and the behaviors that lead to them) should be avoided at all costs. With restraint and discipline, much can be done to prevent finger injuries from occurring in the first place, and that should be every climber's strategy going-in. When injuries do occur, there are some deliberate approaches that can be used for rehabilitation. For more information on this topic, consult Hochholzer & Schoeffl's *One Move Too Many...*, which is an excellent resource for preventing, diagnosing, and rehabilitating climbing injuries that should be on every climber's bookshelf.¹⁰

"An ounce of prevention is worth a pound of cure." — Benjamin Franklin

PLANNING FOR ADEQUATE REST

The first rule of avoiding injury, which every athlete knows (but struggles to follow), is to ensure adequate rest. A significant advantage of the Rock Prodigy program is that rest is built in. The duty cycle for workouts is relatively conservative, because the overarching goal is long-term improvement over a career. Injuries are devastating to such a goal, so extra rest is built in to reduce the risk of overuse injuries. The onus is on the climber to follow the plan, avoid sneaking in extra workouts (like impromptu bouldering sessions at the gym, etc.), and back off when the body signals that additional rest is required.

TAKING CUES FROM THE BODY

While resting adequately will go a long way toward preventing injury, it is still essential for every climber to constantly assess the health of his or her body. One of the primary benefits of progressive resistance training is that it allows an athlete to push right up to her limit without exceeding it. However, that only works if the athlete pays attention to the body's response and adjusts training volume accordingly. While performing exercises, it is critical to pay close attention to feedback from the body, in order to avoid traumat-

ic or *acute* injuries. Hangboard, Campus Board, and Limit Bouldering exercises are the most likely to cause acute injury. For example, if a particular grip position on the hangboard feels too "tweaky," reduce the load applied, or better yet, skip the remaining reps/sets of that grip and move on. Take note of any such sensations so that the load can be reduced for the following workout.

Overuse injuries tend to develop slowly, so are harder to detect, but they can be just as debilitating as acute injuries. These are a result of the accumulation of repetitive microtraumas that occur over the course of repeated training or climbing. The progressive, near-limit training stimulus that eventually forces adaptation first causes this microtrauma (an important part of the super-compensation process). Overuse injuries can only be avoided if the body is able to heal these microtraumas sufficiently between each workout. This means taking any actions possible to encourage the healing process and allowing enough rest for healing, recovery, and super-compensation to occur. Varying the training stimulus from phase to phase also helps prevent overuse injuries by altering the repetitive stress before microtraumas can accumulate into a full-fledged injury.



Repetitive stress to the finger joints, such as that caused by pocket climbing, crimping, and fingerlocking, can culminate in severe overuse injuries. Sonnie Trotter taking a bite out of Stingray, 5.14a, Joshua Tree. © Andrew Burr

Another good preventative measure is to ensure that your training tools are ergonomic. Such tools should align with the body's natural geometry. Any tool that forces the body into an awkward position, places torsional stress on a joint, or otherwise feels orthopedically uncomfortable should be banished. Hangboards are notorious for their poor ergonomics, forcing fingers into contorted positions or placing unnecessary strain on the shoulders. Climbers come in different sizes, while training tools often do not. Select properly sized tools, or make adjustments as needed. A small amount of effort up front preparing ergonomic training tools can save huge amounts of effort on rehabilitation.

If detected early enough and acted upon, overuse injuries can often be sufficiently managed such that the climber can continue training and climbing through the injury. Therefore, every climber should be highly motivated to listen to cues from the body and take action when any hint of injury develops. Specifically, climbers should watch for:

- Stiff/achy joints
- Swollen joints
- Acute pain in any climbing-specific joint (whether caused by climbing or not)
- Reduced joint flexibility/range of motion
- Unusual lumps or protrusions
- Intermittent twinges
- Tingling sensations in fingers, arms, elbows, or shoulders

Only YOU Can Prevent Repetitive Stress Injuries

Climbers often see chronic injuries approaching from a long way off. Yet they watch, transfixed as though observing a slow-motion train wreck, as their elbows/shoulders/fingers get worse and worse, failing to take any meaningful action. Our passion for climbing is so great that we can't accurately judge the obvious trade-off between a few weeks of preventative rest today in place of several months of rehabilitation down the road. Detecting oncoming injuries is less than half the battle; wisdom and action are necessary to make use of that foresight. Adjust training or climbing behavior at the first sign of distress, when small adjustments can pay huge dividends. Once the nagging ache has blossomed into a full-blown debilitating injury, far more effort (and time) will be required to correct the problem.

Climbers should not routinely take anti-inflammatory or pain-relief medications, because they mask these symptoms, allowing minor injuries to develop undetected into major injuries (in fact, research shows "Non-Steroidal Anti Inflammatory Drugs" (NSAIDs) actually hamper soft-tissue healing!).⁹ If any of these symptoms arise, follow the measures suggested in the "Injury Prevention" and "Rehabilitation" sections that follow. If these do not control the injury, increase the rest intervals between workouts and see a doctor.

"Anybody can get strong. The trick is to get strong and not get injured!"

— Wolfgang Gullich¹⁶

INJURY PREVENTION

Besides using a progressive training approach with adequate recovery, preventive maintenance is the next best way to avoid injury. There are several ways to encourage the healing of microtrauma between workouts. These include:

- Eliminating inactive joint stress
- Stretching
- Massage
- Icing
- Preventive bracing/taping

Inactive joint stress occurs when arms, legs, and/or fingers are placed in stressful orientations when not being used. Typically, this involves bending joints unnecessarily, to the limit of their natural range of motion, and maintaining the position for lengthy periods. Some examples of this include:

Elbows and wrists: Sleeping on top of bent arms (as in the fetal position) places unnecessary stress on the elbows (a common injury site for climbers). A climber dealing with *climber's elbow* could reduce this unnecessary stress by sleeping flat on his back with arms straight at his sides.

Back: Poor posture, especially while seated for long periods (while driving, watching TV, or at the office), can be harmful for the back muscles and spinal column. Sit up straight!

Fingers: Avoid clenching the fingers in a fist for long periods (sleeping especially), or other twisting motions.

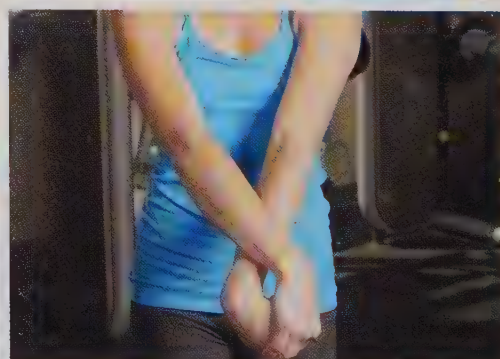
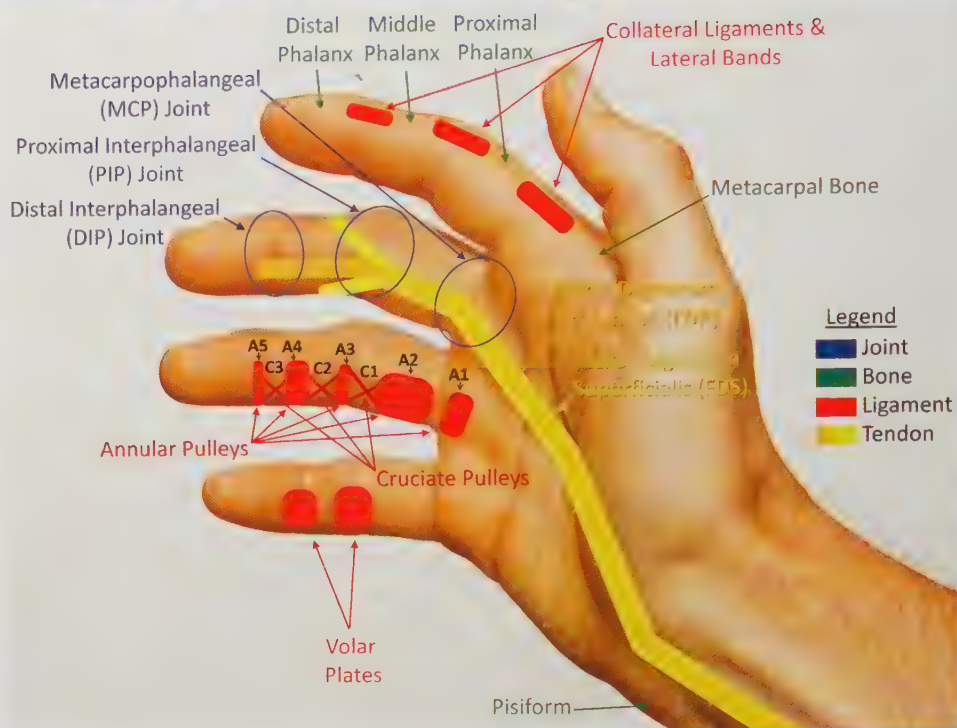
Knees: Avoid squatting on bent knees, crossing the legs, or curling the feet under the chair when seated, which stresses the knee ligaments.

REST, INJURY PREVENTION, AND REHABILITATION

Shoulders: Sleeping on either side places substantial strain on the rotator cuff of the bottom shoulder and should be avoided. This is a common problem among big-wall climbers, who like to spoon on portaledge.

Many of the injuries that can be attributed to “muscle/strength imbalance,” such as *lateral* and *medial epicondylitis* (climber’s elbow), may actually be a result of a general lack of muscle flexibility, or an *imbalance* of flexibility. A tight, inflexible muscle places constant stress on the points where the muscle’s tendon attaches to the bone. Though these attachments are capable of withstanding tremendous loads, they are not intended to be loaded around the clock. A forearm muscle that is tight as a bowstring is constantly tugging on these attachment points, af-

fording little or no rest to the weary tendon. Simply stretching the forearm flexors *and* *extensors* regularly, and after training, can loosen these muscles and relieve this constant stress. Some useful stretches for climbers are shown in the Figures below. Stretch the joint up to a slight feeling of discomfort, or tightness, but not pain. Stretches should last 30 to 60 seconds.



The forearm stretches illustrated here can help prevent elbow inflammation.

How to Beat Climber's Elbow

Climber's elbow (which includes both *medial* and *lateral epicondylitis*) is a very common overuse injury that can result from many climbing activities. It is rarely the result of an acute trauma; instead it usually develops gradually, making it hard to identify the root cause. If not corrected immediately, climber's elbow can develop into a debilitating *chronic* injury (consistent pain lasting more than three months). Chronic injuries require much more intensive rehabilitation and should be avoided. Treatment of such chronic injuries often includes re-creating a controlled inflammation sufficient to trigger the body's internal healing process. It is much easier to identify and treat these injuries at their first sign.

In some cases, these injuries are caused by a flexibility imbalance that results from neglect of the forearm muscles and elbows; in particular, most climbers simply don't know how to stretch their finger extensors, or even that they need to. I suffered from these nagging injuries for many years, but in 2006 I instituted a regime of stretching, massage, and icing that completely eliminated this injury. Occasionally I'll have a flare-up when I get lax with my stretching routine, but it's easily corrected with more diligence. Here's what I do to control the effects of climbing stress on my elbows:

Stretch: Stretch the forearm flexors and extensors thoroughly every day (whether I climbed or not, but especially after climbing workouts). See the Figure on the previous page for recommended stretches, which should include individually stretching each finger's flexor and extensor muscle.

Massage: Massage the forearm flexors and extensors frequently to keep them loose. Simply press into the muscles with the thumb and/or forefingers, and "knead" the muscles. This can be done while watching TV, hiking to the crag, or meeting in the board room.

Ice: Ice the elbows by submerging the entire forearm in an ice-water bath for 10 minutes after any hard workout. Icing should be done last, after cooling down, stretching, etc. (generally one hour or more after exercising).

— Mike Anderson



Icing is a tried-and-true method for reducing inflammation after injuries, but it also aids in recovery from the micro-trauma of everyday training.

Massage plays a role in injury prevention similar to stretching, and generally is used to increase muscle flexibility and thus reduce stress on the tendon's attachment points. However, in addition to this, massage promotes blood flow to the affected muscles, delivering nutrients and removing waste products, thus improving muscular recovery between training or climbing days. While all muscles of the body will benefit from massage, climbers should make a point to massage the forearm flexors and extensors for 10 to 20 minutes within a few hours after any training session or climbing day. This massage should be repeated on rest days as well. In addition, very light massage of the fingers themselves can help promote blood flow to the finger joints (ligaments and tendons), which ordinarily receive very little blood flow. Many climbers find it beneficial to massage their forearms between climbing attempts at the crag, which seems to encourage short-term recovery between burns.

The three previously listed steps for injury prevention —eliminating inactive joint stress, stretching, and massage — all relate to eliminating unnecessary stress on the joints. Icing seeks to prevent injury by encouraging healing between climbing or training events. As Hochholzer & Schoeffl point out in *One Move Too Many...*, "invented over 4 billion years ago, this frozen combination of hydrogen and oxygen atoms has not yet been improved upon."¹⁰ Though the chemical composition is irrelevant (the temperature is the key), they make a good point. Simply cooling an ailing joint does far more for rehabilitation than any manmade cure. Icing reduces swelling and temporarily restricts blood flow, which is beneficial for acute injuries. On the other hand, when the cold is removed, blood flow increases (called *reactive hyperemia*), which is especially beneficial for fingers, which suffer from lower-than-average blood flow, and for flushing tired forearms after a long day of pumpy climbing.

The virtues of icing after injury are well-known, but the scientific community is undecided on its effectiveness for general recovery from training.^{2, 4, 15, 17} Nevertheless, many athletes swear by it, and anecdotal evidence supports its use, especially after very pump-y climbing. Icing can be done after every limit climbing day or intense climbing workout. The best method is to submerge your entire forearm in an ice bath (a tub of water filled with ice), from your fingers up to, and including, your elbow. Try to leave the fingertips out of the water to prevent frostbite and any softening of the fingertip pads. Ice each arm for 10 minutes, and then elevate each arm for another five to 10 minutes. Only ice well-after completing a workout, after performing the appropriate cooldown or supplemental exercises, and after stretching and massage. Stretching a muscle immediately after icing can cause injury! A good rule of thumb is to wait at least one hour following an intense workout (not including the cooldown) before icing.

Finally, some acute and overuse injuries can be avoided with preventive taping before training or climbing.¹² Tape is generally applied to the fingers to support the *pulleys*, which are ligaments that hold the flexor tendons in place and allow joint movement. These are often injured by overloading while in a crimp grip. The photo below shows a common method for supporting the A2 and A3 pulleys with tape. Tape must be very tight to be effective, and should be replaced between every burn, because it stretches. In general, taping should be avoided during training so that the finger ligaments are subjected to stress in a controlled setting (unless dealing with an existing injury). This allows the support structure of the fingers to strengthen, reducing the likelihood of injury in the future. On the other hand, it is not a bad idea to tape up for activities that will place an unknown and uncontrolled load on the fingers, such as dynamic, powerful bouldering at a new area, or the first few attempts on a difficult project.



H-Taping has been shown to reduce crimping stress on the A2 and A3 pulleys. Tape must be tight and replaced regularly to be effective.

“Going through the moves with confidence is difficult because you’re just not sure when it’s going to be too much. You don’t know exactly what your fingers can take, and when you go too far, it’s really bad.” — BJ Tilden⁸

REHABILITATION

As discussed in the accompanying sidebar, “When to See a Doctor,” the vast majority of climbing injuries are best treated with diligent *rehabilitation*.^{13, 14} Rehabilitation (also known as physical therapy) is a process of restoring damaged tissue through methodical and controlled application of stress. If this sounds similar to progressive overload, that’s because it is. The primary difference is that the loads involved must be vastly reduced during the initial stages of rehab. Furthermore, a rehabilitating athlete may require more rest than a typical healthy athlete in training. Rehab is typically not fun, but it is an extremely powerful method of injury treatment.

For climbers who train, rehab becomes a relatively simple matter. The tools and methods used for improving strength in healthy joints can be easily applied to injured joints. Of course, *this only works if the resistance applied is drastically reduced!* This is critically important and bears repeating. Initially, the resistance may be so low that assistance (from a partner

Shouldering the Burden

Shoulders are also susceptible to chronic injury caused by an imbalance of flexibility, poor posture, and poor technique when reaching. To avoid such injuries, stretch the pectoral muscles, stand up straight, and avoid the hunched-over, caved-in posture typical of many muscle-bound climbers. As noted earlier, avoid sleeping awkwardly or otherwise unnecessarily stressing the shoulders. In severe cases, a physical therapist (“PT”) can help restore correct posture and joint motion. A PT will be able to prescribe numerous preventative maintenance exercises for the shoulders that use very low intensities while traversing the joints’ range of motion. The supplemental exercises listed for shoulders in **Chapter 6: Strength** can also be used as a starting point to develop injury-resistant shoulders — just be sure not to push too hard. If experiencing nagging night pain in one or both shoulders, see a PT, as this is a symptom of a torn rotator cuff.

When to See a Doctor

Taking a climbing injury to a doctor can be a lot like taking a faulty iPad™ to a plumber. Most doctors simply have no experience with such injuries. It's not that doctors are *bad* per se; it's just that general practitioners are not trained to treat sports injuries. While an orthopedist (a doctor specializing in injuries to the skeletal-muscular system) may help diagnose them, sports injuries require rehabilitation to treat, and for that, you need a sports physical therapist. Doctors are trained to do one of two things: prescribe drugs and perform surgery. If you think one of those approaches will help you, a doctor might be just the right person to visit (note: those interventions are *rarely* helpful!).^{13, 14}

There are many skilled, caring doctors who can help with invasive treatments. If you do decide to visit a doctor, see a specialist who focuses on your area of injury (for example, see a hand specialist for finger injuries, but keep in mind, a hand *surgeon* may be more biased toward performing *surgery*). Seek referrals from other climbers when possible. Arm yourself with information and be prepared to coach your doctor about climbing, the stresses involved, potential diagnoses, and recommended treatments. Bring a copy of Hochholzer & Schoeffl's *One Move Too Many...* and use it as a visual aid.¹⁰ It's up to you to make the visit valuable.

One good reason to see a doctor for a typical climbing injury is that doctors have access to diagnostic tools like ultrasound, x-rays, and MRIs. If you have a nagging injury, and rehab is going slow, you might consider seeing a doctor in order to use one of these tools. Keep in mind, however, that the treatment is unlikely to change much, regardless of the diagnostic results. The primary value of the diagnosis is to rule out surgery. Another treatment doctors can offer is *cortisone*. Cortisone is a steroidal drug that can be injected into inflamed tissue (such as climber's elbows) to reduce inflammation and numb pain. While it is proven to reduce pain in the short term, recent research suggests it has no long-term benefits and may even be harmful, weakening connective tissue (if considering a cortisone injection, be sure to discuss these findings with your practitioner).³

At the end of the day, rehabilitation is the solution to the vast majority of climbing injuries.^{13, 14} As such, the logical course of action is to start with rest and rehabilitation, then visit a doctor only if several weeks of these measures fail to produce results. In the case of severe acute trauma, where you feel surgery might be necessary, see a doctor immediately, but come prepared and take charge of your diagnosis and treatment.

or healthy limb) is required just to perform weightless range-of-motion (ROM) exercises. Climbers must listen to their bodies to determine how much resistance to use, and when or how much to increase that resistance. During rehab, the consequences of using too much resistance are much greater, so caution and patience are vital.

When rehabilitating chronic overuse injuries, it's important NOT to use NSAIDs like Ibuprofen or aspirin. First off, very high doses are required to achieve any anti-inflammatory effect (around 800mg every four hours). Second, these drugs interfere with the body's natural response to inflammation, hampering recovery.

STAGES OF INJURY REHABILITATION

Nearly all injuries will follow the same basic rehabilitation approach. Even in extreme cases requiring surgery or other invasive treatments, rehab is vital to complete recovery. The training tools and methods used may vary depending on the site and type of injury, but the following general stages of rehab apply to the vast majority of climbing injuries:

1. **Rest:** The purpose of this stage is to eliminate the source of the injury, reduce swelling, and allow any initial pain to subside before transitioning to more active recovery measures. Ice the injured area several times per day for the first three days following injury and also avoid heat; then, continue to ice after each rehab training session. Take advantage of any downtime by analyzing the events that led to the injury in order to avoid repeating any mistakes you made. Consider that acute injuries are often the result of several weeks or months of repetitive abuse. Finally, it's important to realize that little healing occurs during the rest stages. Many climbers rest too long after injury, then after many weeks of inactivity they are surprised to discover no improvement when they return to activity. Injured tissue requires frequent stimulation to promote healing, which can be as simple as pain-free stretching and range-of-motion exercises (more to follow). Rest periods can be as short as one week, but generally not more than four.

2. Alternative Training: The purpose of this stage is to find a productive use for the climber's time and energy that contributes to his long-term goals. A climber with an ankle injury might hangboard to improve finger strength, while a climber with inflamed elbows might toprope technical slabs to improve footwork. This type of training can be quite motivating. Even though one aspect of fitness has regressed, other aspects can be improving, suggesting that once recovered, the climber will be that much better than he was before the injury. Depending on the injury, this stage can begin concurrently with the rest stage and should continue until the climber is healthy enough to resume normal training activities.

3. Range-of-Motion (ROM) Exercises: Once the initial pain of injury has subsided, begin moving the joint within the available pain-free range of motion. In addition to restoring flexibility, these exercises will improve blood flow and stimulate recovery. Start by warming the injured joint (preferably through light exercise, but heating pads or warm water work too). Begin by exercising the joint within the comfortable range, gradually increasing the ROM as the joint warms up. Include some very light stretching once the joint is completely warm. It can take many weeks or even months to completely restore ROM, since any residual swelling will inhibit joint movement.

As more aggressive rehab activities are added, swelling may return, further hampering ROM. Be patient and continue using ROM exercises throughout the entire rehab process.

4. Progressive Rehab: Once major swelling has subsided and basic ROM is restored, it's time to accelerate the recovery process. Progressive rehab begins by performing the basic sport-specific functions of the joint with extremely low resistance (for example, no weight other than that of the limb may be an appropriate starting point). Perform progressive rehab just as you would perform progressive overload training, but consider performing more repetitions of each exercise. Push the injured area just beyond the comfort zone, then rest for several days to allow the tissue to recover, and repeat. The primary differences are that rehab resistance should be vastly lower than training resistance, and rest periods between rehab sessions should be greater (begin by taking three full rest days between rehab activities, then adjust based on the body's response). Pay close attention to the body's response to rehab and make adjustments as needed. Just as with progressive training, eventually the body may stop responding to progressive rehab. In these cases, a periodic approach may work. If sufficient recovery has occurred to permit safe climbing, consider a few weeks of mod-

Keep Fingers Looking Good

There are many ways to injure a finger in climbing, but A2 pulley strains and collateral-ligament strains are the most common. I've suffered both on numerous occasions. In 2012, I severely strained the A2 pulley in my left ring finger while working an open project at Shelf Road. This injury was a long time coming — the culmination of several years of intermittent tendonitis. This was the worst finger injury of my career, and at times I seriously doubted I would ever recover completely, writing in my journal: *"I think I really hurt my A2 pulley pretty badly. It's hard to imagine bearing down on a crimp any time in the near future. At this age, you can't help but wonder which injury will end your career. What makes me hopeful is that I've been in this position many times, and it always seems hopeless in the moment, yet I've always managed to come back stronger than ever."*

I began my rehab with two weeks of complete rest, followed by a four-week course of conservative, progressive hangboarding. I began with extremely low weight (half my normal starting load) and took an extra rest day each week. At the end of this stage, my strength had improved considerably but I was nowhere near cured. I took three weeks off, then completed another four-week course of hangboard rehab. At the end of this regimen, the hangboard results indicated that my injured finger was at about 80 percent of its previous strength, so I spent the following month climbing moderates before beginning a third round of hangboard rehab. By the end of the third course, I had set new personal bests on the hangboard and was climbing as well as ever. The entire rehab process took six months, including several setbacks, and plenty of hard work. Still, it was well worth the effort: I am now 100 percent pain free, and considerably stronger than I was when the injury occurred. In February 2013, I returned to Shelf to complete the first ascent of Flight of the Phoenix, the route that pushed my finger past the brink.

— Mark Anderson

erate climbing before returning to more deliberate progressive rehab. Severe injuries can require three or more iterations of progressive rehab spanning several months of intensive work.

- 5. Progressive Training:** The final stage of rehab is to return to normal training activities, albeit at a reduced volume initially. Again, additional rest may be required between training sessions, and you should expect some mild swelling to return following intense workouts. Continue to ice the injured area after training, and complete ROM exercises regularly. This stage will culminate in a return to climbing. Consider the circumstances that initially led to injury and strive to correct them going forward.

PSYCHOLOGICAL CONSEQUENCES OF INJURY

Injuries can feel overwhelming and highly demoralizing at the onset. In extreme cases, climbers end a salvageable career simply because the rehab process is too overwhelming. Therefore, a positive mental attitude and faith in the recovery process are essential to effective rehabilitation. It will require a great deal of time, hard work, and may involve a setback or two, but with perseverance, complete recovery is usually possible. Once injured, many athletes who diligently rehab come back to their sport even stronger than before. Use the rehab period as a time to re-focus

on other physical weaknesses or improve movement skills.

While no climber would opt for a lengthy rehab, it's not all bad. Rehab can provide a good opportunity to explore other interests. Find something to put energy into that will distract from the disappointment of not climbing. Use these activities to replenish motivation, and then apply that energy to rehab.

Patience is critical to the rehab process; realize that rehab will take time and hard work. Expecting quick results often causes climbers to progress too quickly, risking relapse and a longer rehab period. Remain cognizant of long-term climbing goals; these can be powerful motivators when facing the long road to recovery. In the long run, the best way to overcome injury is to strive for 100 percent recovery and to achieve that state as quickly as possible. This may seem painfully obvious, but few climbers pursue this aim. The goal of achieving 100 percent recovery means a climber should not accept limping through life with a nagging injury that is constantly holding him or her back (or repeatedly relapsing). The best way to achieve recovery as quickly as possible is through conservative, steady (and therefore patient) progress. Pushing slightly too hard usually results in a setback that costs the climber many weeks of additional recovery and rehab. Go slow, consider the long view, and don't quit until complete recovery is achieved.

SUMMARY

- Resting is the final phase of the Rock Prodigy periodic training program. Intermittent rest days are also critical within the training and performance phases.
- Rest days are essential to allow recovery and super-compensation to take place during the progressive training process. There are numerous supporting activities that should be performed on rest days including skin care, supplemental exercises, preventative maintenance exercises, skill practice, and planning.
- Skin is the climber's connection to the rock, and caring for it is critical. The best way to avoid skin injuries is to prepare the skin for climbing in advance, and to use your skin judiciously.
- The Rest Phase is necessary for continued long-term improvement. The decision of when to start the Rest Phase should be based on past training data, signs of injury, or declining performance. Never climb beyond a six-week Performance Phase.
- It is much easier to prevent climbing injuries than it is to recover from them. Rest, preventative maintenance exercises, and monitoring the body's response to training and climbing are the best ways to avoid injuries.
- Injury rehabilitation follows a process similar to progressive training but uses exercises of a much lower intensity, with more repetitions and more rest. Mental toughness and perseverance are needed to see things through, but complete recovery is nearly always possible.